

Important Notes Of MySQL Part 2

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Constraints

There are 6 types of constraints
in MySQL

1st one is Primary key,

2nd one is Null.

Not null

Foreign key,

check clause

Default

query

waq where you will create a table called ggemploye which will have the following constraints, Empno primary key, ename cannot be left blank and basic salary has to be minimum Rs. 1,400/- and city default 'Mumbai'

```
mysql> create table hsemployee
      (empno numeric(5) primary
key,
      ename char(20) Not Null,
      doj      date,
      basic numeric(9,2)
check(basic >=1400),
      city      varchar(19) Default
'Mumbai');
```

Table created.

While trying to add records in
following scenarios

```
mysql> insert into hsemployee  
      values(1004, null,  
sysdate(), 8999, Default);
```

```
ERROR 1048 (23000): Column  
'ename' cannot be null
```


mysql> insert into hsemployee
 values(1004, 'DinDayal',
sysdate(), 1200, Default);

```
ERROR 3819 (HY000): Check  
constraint 'hsemployee_chk_1' is
```

violated.


```
mysql> insert into hsemployee  
      values(1004, 'DinDayal',  
sysdate(), 1450, 'London');
```

1 row created.

=====
=====
=====

Functions or Group by Functions

Group by functions are also known
as aggregate functions or
numeric functions.

There are only 5 functions in
that.

Sum, max, min, avg, count();

Sum() : - it is used for
addition purposes.

```
mySQL> select sum(comm) from  
salespeople;
```

SUM(COMM)

83970.22

```
mySQL> select sum(comm) Incentive  
from salespeople;
```

INCENTIVE

83970.22

```
mySQL> select sum(comm)  
"Incentive Amount" from  
salespeople;
```

Insentive Amount

83970.22

Max() : It is used for printing
highest values.

mySQL> select max(comm) from
salespeople;

MAX(COMM)

39393.33

Min() : it is used for printing lowest values.

```
mysql> select min(comm) from  
salespeople;
```

MIN(COMM)

355

Avg() : It is used for printing average values.

```
mysql> select avg(comm) from  
salespeople;
```

AVG(COMM)

8397.022

count() : it is used for
counting for no of records in the
table.

mySQL> select count(*) from
salespeople;

COUNT(*)

20

```
*****  
*****  
*****
```

IN Clause

```
*****
```

In Clause works faster for fetching records and when table has huge database.

It can be used with all char, varchar2, date and number attributes.

waq where you will print all details of snum 1001, 1008, 1004, 1090

```
mySQL> select snum, sname, city  
        from salespeople  
        where snum in (1001,  
1008, 1004, 1090);
```

```
SNUM SNAME
```

CITY

1001 Kalia
Navi Mumbai
1090 Dr. Jun Jun Wala
London
1008 James Bond
Mumbai

waq where u will print all
salesperson residing in London or
newyork or chicago or mumbai.

```
mySQL> select snum, sname, city
        from salespeople
        where city in
('London', 'New York',
'Chichago', 'Mumbai');
```

SNUM SNAME
CITY

1090 Dr. Jun Jun Wala
London

1008 James Bond
Mumbai

1400 Dr. Rahul
London

8977 Shri amit
London

1577 John dikosta
Mumbai

1013 Dr. Batli Wala
Mumbai

1678 Dr. Praveen K
London

9393 Dr. Batuk Patel
Mumbai

1009 Kishan JunJun wala

London
1409 Shivaji Patil
Mumbai
9001 James Singh
Mumbai

SNUM SNAME
CITY

9009 James Patel
London

12 rows selected.

Waq where you will display snum,
sname and city of salesperson not
residing in
London, newyork or chicago or

mumbai.

```
mySQL> select snum, sname, city
        from salespeople
        where city not in
('London', 'New York',
'Chichago', 'Mumbai');
```

SNUM	SNAME	CITY
1001	Kalia	Navi Mumbai
1234	seema	bihar
1456	Ranjit singh	Jaipur
1040	Rana Pratap	Los Angeles
1666	Dr. Lambu	Jumritalaya

1899 Dr. Sokhi
Hyderabad
5779 raul
hyderabad
444 dkdkd
dkdkd
49494 shankara
Sydney
1011 Dr. Preran K
FrankFurt
1089 Dr Navya
Johanes Burg

SNUM SNAME
CITY

4433 Rahul
Sydney
3453 Hari Bhajan
Hyderabad
1982 SUkwinder sokhi

Patila

1403 Dr. Shriman Lagoo

Sydney

15 rows selected.

waq to print snum, sname, and
city of salespeople whose sales
number should not
be 1008, 1001, 1004 and 1090

```
mySQL> select snum, sname, city
        from salespeople
        where snum not
in (1001, 1008, 1004, 1090);
```


Important string Functions

Upper and lower functions in the same query.

```
mysql> select upper(sname) name,  
lower(city) city, comm  
        from salespeople;
```

NAME	CITY	COMM
KALIA	london	
DR. JUN JUN WALA	london	1810.14
JAMES BOND	mumbai	
SEEMA	bihar	9800.14

RANJIT SINGH		jaipur
	1701.14	

DR. RAHUL		london
	9579	

RANA PRATAP		los
angeles	3578.14	

DR. LAMBU		
jumritalaya		
DR. SOKHI		
hyderabad		
RAUL		
hyderabad		4934

NAME	COMM	CITY

DKDKD		dkdkd
SHRI AMIT		london
SHANKARA		sydney
DR. PRERAN K		
frankfurt		9494

DR NAVYA		johanes
burg		

JOHN DIKOSTA		mumbai
--------------	--	--------

355

RAHUL		sydney
-------	--	--------

39393.33

HARI BHAJAN

hyderabad	3325.33
-----------	---------

DR. BATLI WALA	mumbai
----------------	--------

RAHULD MODAK

DR. PRAVEEN K	london
---------------	--------

3334.22

NAME

CITY

COMM

LUCY KHANNA

johanesburg	353
-------------	-----

DR. BATUK PATEL	mumbai
-----------------	--------

333

SUKWINDER SOKHI	patila
-----------------	--------

393

KISHAN JUNJUN WALA

london

26 rows selected.

lpad() Function

```
mySQL> select lpad('Rama was a  
great king ', 72, '*') lpad  
from dual;
```


-

RPAD()

```
mySQL> select rpad('Rama was a  
great king ', 72, '*') rpad  
from dual;
```

RPAD

Rama was a great king

-

-

Ltrim() : will remove the left
trailing blank spaces from the
string.

mySQL> select ltrim(' Preran is the V.C of Bangalore University')

```
' ) from dual;
```

```
LTRIM( 'PRERANISTHEV.COFBANGALOREU  
NIVERSITY
```

```
-----
```

```
-----
```

```
Preran is the V.C of Bangalore  
University
```

```
-----
```

```
-----
```

```
-----
```

```
-
```

```
Rtrim( ) :-
```

```
*****
```

```
mhySQL> select rtrim('
Preran is the V.C of Bangalore
University  ')
          from dual;
```

```
RTRIM( 'PRERANISTHEV.COFBANGALOREU
```

NIVERSITY')

Preran is the V.C of
Bangalore University

-

Length() : -

```
mysql> select length(' India wins  
world cup of football in 2040  
' )
```

```
from dual;
```

```
LENGTH( ' INDIAWINSWORLDCUPOFFOOTBA  
LLIN2040' )
```


-

```
mysql> select  
length(786868767868767868768768)  
from dual;
```

```
LENGTH(786868767868767868768768)
```

24

-

```
mysql> select length(' Jaipur is  
a nice city      ') from dual;
```

```
LENGTH(' JAIPURISANICECITY')
```

```
-----  
-----  
-----  
- 28
```

```
mySQL> select length(trim(' Jaipur is a nice city '))  
from dual;
```

```
LENGTH(TRIM(' JAIPURISANICECITY'))  
-----  
21
```

```
-----  
-----  
-----  
-
```

```
MySQL> select trim(' Jaipur is a  
nice city ') from dual;
```

```
TRIM(' JAIPURISANICECI  
-----
```

Jaipur is a nice city

-

```
mySQL> select substr('
Sindhurg is a nice fort ', 5,
8) substr
      from dual;
```

SUBSTR

indhudur

-

```
mySQL> select substr('
Sindhurg is a nice fort ', 75,
8) substr
```

```
from dual;
```

```
-----  
-----  
-----  
-
```

```
mySQL> select substr('Sindhudurg is a nice fort ', 5,  
88) substr  
from dual;
```

SUBSTR

```
-----
```

```
indhudurg is a nice fort
```

```
-----  
-----  
-----  
-  
-----  
-----  
-----
```

-

1)Alter Table.

The ALTER TABLE Statement

The ALTER TABLE statement is used to add, delete, or modify columns in an existing table.

SQL ALTER TABLE Syntax

To add a column in a table, use the following syntax:

```
ALTER TABLE table_name  
ADD column_name datatype
```

```
mysql>ALTER TABLE salespeople  
        ADD DateofJoin Date;
```

-

The above query will add a column
to the table employee;

To drop a column in a table

```
mysql>ALTER TABLE salespeople  
        DROP COLUMN DateOfjoin
```


-

to modify a column in a table
query is

```
mysql>alter table salespeople  
        modify comm numeric(12,2);
```


uniquee key.
Indexing